Federated Ophthalmic Disease Diagnosis Based on Weighted Averaging and Local Optimization

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Abstract*—Deep learning and computer vision (CV) can easily recognize pictures that the human doctor would have to spend a lot of time and labor to process. Thus, they have a wide range of applications in the field of ophthalmology, which is highly dependent on fundus images. However, training a robust and highly accurate model requires a large number of labeled high-quality images, which is often difficult for a single institution to provide. A natural solution is to use data from multiple medical centers to train the model. But due to privacy protection, data heterogeneity, and other problems, these data cannot be shared directly. Based on a real case of ophthalmic disease diagnosis with multiple data sources, this paper proposes solutions to *the heterogeneity of data amount and features* between datasets, establishes a heterogeneous federated learning framework, and applies federated learning to the diagnosis of multiple types of ophthalmic diseases for the first time. The experimental results show that federated learning can greatly improve the diagnostic accuracy of clients with less data, and the accuracy of clients with more data can also be slightly improved. The ablation experiment has demonstrated the effectiveness of the proposed method for handling data heterogeneity. Our research shows that through heterogeneous federated learning, we can use multi-source, non-independent-identical data to build accurate disease diagnosis models, and make clients who cannot obtain effective models get a practical level disease diagnosis model.

Keywords—Ophthalmic Disease Diagnosis, Federated Learning, Data Heterogeneity

I. INTRODUCTION

Ophthalmology heavily relies on huge amounts of fundus images, and processing these images by human doctors could cost plenty of labor and time. Thus, deep learning and computer vision have been widely used in the field of Ophthalmology [1]. Based on different purposes, the application of deep learning in ophthalmology can commonly be divided into several tasks, including lesion detection [2], lesion segmentation [3], biomarker segmentation [4], disease diagnosis [5], disease grading [6], image synthesis [7], etc.

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All of these tasks cannot be completed without the sponsor of the ophthalmic database. However, compared with the traditional deep learning benchmark dataset, the medical dataset often has the problems of less data [8] and unbalanced data amount between categories. Therefore, the accuracy of the deep learning model trained on an ophthalmic database is often quite low and its generalization is poor. A natural solution is to gather databases owned by different institutions into a comprehensive data pool, based on which the model can be better trained. Unfortunately, due to the requirement of protecting patient privacy, most ophthalmic databases are not accessible to the public.

Fortunately, federated learning proposed by [9] offers an effective solution to the dilemma between privacy protection and model performance. This kind of deep/machine learning paradigm requires every participant client to train a model separately and locally, therefore guaranteeing that the data from each client will not be delivered to others. The collaboration part is realized by aggregating model parameters or gradients that are updated to the cloud server, and then clients use the aggregated model to continue training. In the ideal case, the aggregated model can extract knowledge from all clients, so each client model will have better performance and faster training speed than the one training alone. Federated learning plays a more and more important role in the field of data-driven medical deep learning [10-13], which attaches great importance to privacy protection.

II. METHODOLOGY

A. Problems of data heterogeneity

The concept of data heterogeneity has many different explanations in past research. In this article, we mainly focus on two aspects: *data amount heterogeneity* and *feature heterogeneity*, which are commonly seen in ophthalmic datasets and more suitable for our application scenario.

1) Data Amount Heterogeneity

Due to different hospital scales and patient amounts, the data quantity of different databases usually differs greatly. Models trained with less data typically have lower classification accuracy, so when aggregated with models trained with more data, these models can reduce the accuracy

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of the aggregated model. Many medical studies on federated learning have encountered the same problem [14-16].

2) Feature Heterogeneity

Even for the same patient, different hospitals may collect different fundus images, because they may use different instruments, different contrast mediums, or different light sources, which ultimately lead to the difference in image features. In case of *feature heterogeneity*, the generalization of the aggregated model may be destroyed, causing a decrease in model accuracy [17].

These two kinds of data heterogeneity exist among our datasets and have negative effects on the performance of federated learning. Strategies to deal with these problems will be discussed in this work.

B. Heterogeneous Federated learning

Disease diagnosis is a task belonging to the field of image classification, in which CNN and transformer-based models are currently the most commonly used ones. Considering that the data amount of medical datasets is usually much less than that of benchmark datasets like CIFAR-10, and transformer-based models have poor performance on datasets with only a few hundred photos. Recently, Facebook AI Research and U. C. Berkeley [18] proposed a pretty new CNN-based image classification model, ConvNeXt, which is state-of-the-art in many tasks in the CV field and is finally adopted as a local model for each client in this work.

Assume that there are K clients hosting K databases $\mathcal{D}_1, \ldots, \mathcal{D}_{\mathcal{K}}$, with size $n_1, \ldots, n_K$. The total amount of data $n = \sum_{i=1}^{K} n_i$. Disease diagnosis is a typical supervised learning task, and thus let $X \subset \mathbb{R}^{p \times p}$ be an instance space, $Y \subset \mathbb{R}^q$ be a label space, and the ConvNeXt model $f: X \rightarrow Y$ parameterized by θ^f is trained. Given a ground-truth labeling function $c^*: X \rightarrow Y$ and a cross-entropy loss function $D_{KL}: Y \rightarrow R$, the objective function of training f is formulated as (1):

$$\min_{\theta} E_{x \in D} \left[D_{KL} \left(f(x; \theta^f), c^*(x) \right) \right] \tag{1}$$

As we presented in II-A, two kinds of data heterogeneity will be considered in the procedure of federated learning and need to be handled.

1) Weighted averaging

As we all know, data-driven models can achieve better performance with more data. However, due to the requirement of privacy protection, these databases cannot be simply pooled together. To complete the target of synthesizing knowledge from all K datasets, federated learning lets every client train a ConvNeXt model f_i separately, and upload the corresponding parameters θ_i^f to a central server, which will aggregate all the parameters to a set of centralized ones θ_{ctr}^f. The strategy of weighted averaging is used here:

$$\theta_{ctr}^f = \sum_{i=1}^{K} \pi_i \times \theta_i^f \tag{2}$$

where $\pi_i = \frac{n_i}{n}$ is the aggregating weight computed based on the data amount of each client. The model trained with a smaller amount of data usually has worse effects, thus it should make less contribution to the aggregated model. Then each client downloads θ_{ctr}^f from the server to the local model and continues to train it.

2) Local optimization

When tackling *feature heterogeneity*, in addition to little tricks like regularizing images from different clients into the same format (size, channel, and color space), we also draw lessons from transfer learning that instead of expecting a model that is suitable for every dataset, we fine-tune the aggregated model locally to obtain K best models corresponding to K datasets.

The overall objective function of the global optimization strategy is (3), which indicates that we want to minimize the sum of loss on each client.

$$\min_{\theta_{ctr}^f} \sum_{i=1}^{K} E_{x \in \mathcal{D}_1} \left[D_{KL} \left(f(x; \theta_{ctr}^f), c^*(x) \right) \right] \tag{3}$$

To deal with the problem of *feature heterogeneity*, local optimization is adopted, which means the models are optimized on their corresponding clients instead of one overall model suitable for all clients. The objective function (3) is altered to (4):

$$\begin{cases} \min_{\theta_1^f} E_{x \in \mathcal{D}_1} \left[D_{KL} \left(f(x; \theta_1^f),\ c^*(x) \right) \right] \\ \vdots \\ \min_{\theta_K^f} E_{x \in \mathcal{D}_{\mathcal{K}}} \left[D_{KL} \left(f(x; \theta_K^f),\ c^*(x) \right) \right] \end{cases} \tag{4}$$

1) Pipeline of heterogeneous federated learning

In summary, a practical and complete heterogeneous federated learning workflow is proposed (see Algorithm 1 and Fig. 1).

Algorithm 1 Workflow of heterogeneous federated learning

Require: Datasets $\mathcal{D}_1, \ldots, \mathcal{D}_{\mathcal{K}}$ and corresponding initialized models $\theta_1^f, \ldots, \theta_K^f$, number of communication rounds m.

if *feature heterogeneity* exists **then**
  Adopt local optimization.
else
  Adopt global optimization.
end if
Generate a public and private key for homomorphic encryption
for $k = 1$ to m **do**
  Each client extracts several layers of the local ConvNeXt model θ_i^f, and encrypts them using homomorphic encryption.
  The unencrypted parts are flattened to keep model structure secret and uploaded to the central server together with the encrypted parts and label information.
  The server computes aggregated model parameters using weighted averaging.
  The aggregated parameters are sent to each client. The encrypted parts get decrypted and assemble a complete model with the unencrypted parts.
  Each client continues to train the local model θ_i^f on $\mathcal{D}_i$.
end for

Given K clients, K ConvNeXt models are established and will be trained on their corresponding local datasets. According to the situation of feature heterogeneity, a global or local optimization strategy will be adopted.

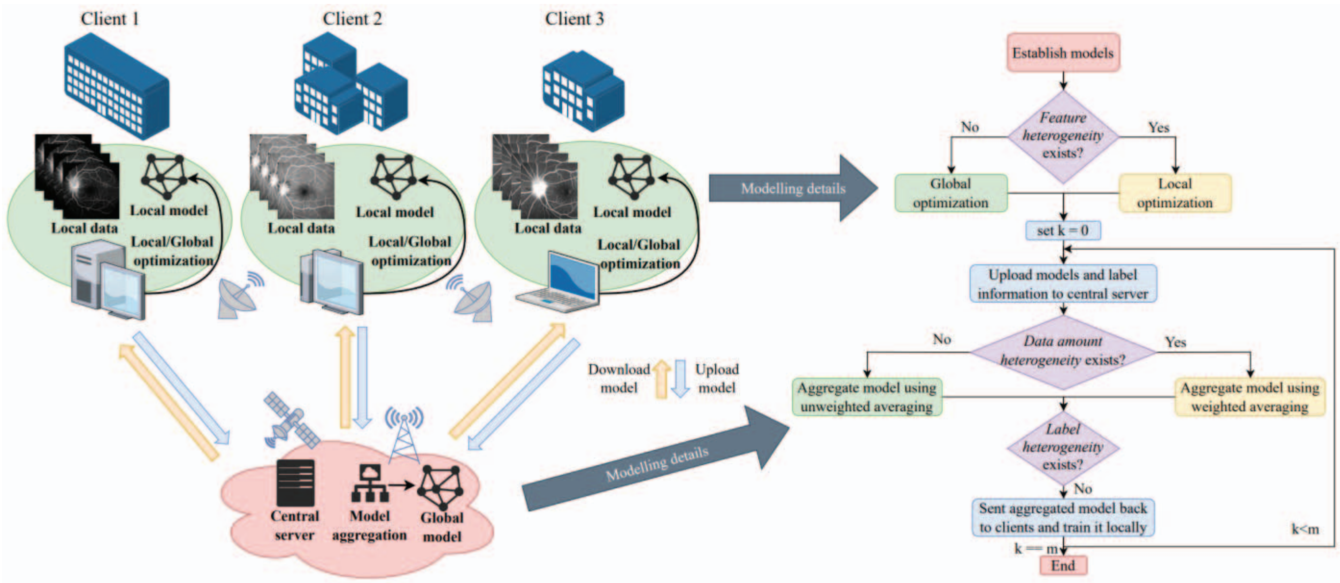

Fig. 1. Overview of the proposed heterogeneous federated learning.

The parameters θ_i^f of each model will be extracted and homomorphically encrypted for more strict privacy, and finally uploaded to the central server after every specific number of epochs. When the server received all the model parameters, it will compute the aggregated model using weighted averaging. After that, the aggregated model will be sent to each client and decrypted. The clients will continue to train on it, then one round of communication is done.

III. Case Study

This case is based on a real ophthalmic disease diagnosis task, which is a suitable application scenario for federated learning and contains three kinds of problems related to data heterogeneity.

A. Dataset description

Even though colorful fundus photography (CFT) and optical coherence tomography (OCT) are more commonly seen as ophthalmic examinations in clinical practice, fundus fluorescein angiography (FFA) serves as the current gold standard for detecting typical pathological changes such as microaneurysms, non-perfusion regions, and vascular leakage. Thus, FFA is a more powerful tool for ophthalmic disease diagnosis [19]. Datasets in this article are also based on FFA images.

There are three adopted FFA datasets, denoted as $\mathcal{D}_1$, $\mathcal{D}_2$, and $\mathcal{D}_3$, which are collected from real hospitals. $\mathcal{D}_1$ and $\mathcal{D}_3$ are collected from Southeast China, while $\mathcal{D}_2$ comes from Northwest China. $\mathcal{D}_1$ has the largest data amount (708 images). $\mathcal{D}_2$ and $\mathcal{D}_3$ have less data, only 150 and 122 images respectively. Therefore, there exists *data amount heterogeneity* among datasets. For each client, we take 80% images as the training dataset and the rest 20% as the validation dataset.

Besides, due to the reasons that we have presented in II-B-2, the FFA images in the three datasets have different feature performances, as shown in Fig. 2.

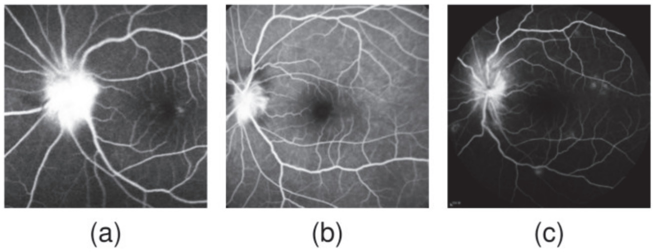

Fig. 2. Three images of the same disease respectively from three datasets show a great difference of modality.

To better illustrate the feature difference between datasets, we use the feature extraction layer of a pre-trained ConvNeXt network to transform every image to a 768-dimension vector, and then use the T-SNE algorithm [20] to reduce the dimension to 2, which is easy to be visualized in a 2-D image. The clustering result is shown in Fig. 3. The x-axis and y-axis are the production of dimensionality reduction and have no actual meaning. Different point types indicate the ground truth client label of the image. Fig. 3 clearly shows that all images from the same dataset are clustered together, confirming the existence of feature heterogeneity. The k-means clustering algorithm is used directly on the 768-dimension vectors to make sure the visualized method is correct, and the same result is obtained.

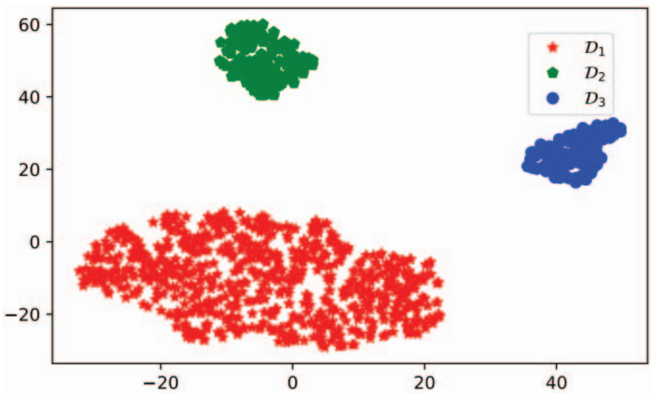

Fig. 3. Clustering result of the three datasets. The x-axis and y-axis are the production of dimensionality reduction while different point types indicate the image's ground-truth client label.

There are 8 classes of disease in each client, which are AION (Anterior Ischemic Optic Neuropathy), AMD (Age-related Macular Degeneration), BRVO (Branch Retinal Vein Occlusion), Coats (Coats' Disease), CRVO (Central Retinal Vein Occlusion), DR (Diabetic Retinopathy), ERM (Expiretinal Membranes), and normal cases. More details are shown in Table I.

TABLE I. DATA DISTRIBUTION IN ALL CLASSES OF 3 DATASETS

Disease Type	Amount of Images in Each Dataset		
	$\mathcal{D}_1$	$\mathcal{D}_2$	$\mathcal{D}_3$
AION	100	20	5
AMD	100	20	18
BRVO	100	20	9
Coats	10	10	0
CRVO	100	20	3
DR	100	20	37
ERM	100	20	14
Normal	98	20	36

B. Experiment Purposes and Settings

Firstly, three models are trained separately on $\mathcal{D}_1$, $\mathcal{D}_2$, and $\mathcal{D}_3$, and they are prepared for comparison with the ones trained under the federated learning paradigm. Then we designed Experiments A to C as below.

1) Experiment A

In this case, we do federated learning to obtain the universally best model, in despite of the feature heterogeneity. When aggregating models in the central server, an unweighted averaging strategy is adopted.

2) Experiment B

In this case, we expect that each client can get the best model corresponding to its own dataset, so local optimization will be adopted, while the average of the models is still unweighted.

3) Experiment C

Both weighted averaging and local optimization strategies will be adopted. After each epoch of local training, the model parameter is sent to the server for aggregation and then downloaded to the local client.

Ablation experiments A, B, and C will be compared with each other to check the effectiveness of the strategies that we proposed to deal with data heterogeneity.

The experiments were repeated 12 times. Taking the average value of the accuracy on the validation set in the last 1/4 training procedure as an evaluation criterion, the highest-scored round and lowest-scored one are discarded, and the results of the remaining 10 rounds are averaged as the valid data.

C. Universal implementation details

The batch size is set to 8, and the total number of training epochs is 300. According to the cosine descent strategy, the learning rate decreases from 0.0005 to 0.00005. The weight of each client when aggregating models is the proportion of the number of images in its datasets to the total amount of images in all client datasets, which is denoted as:

$$w_i = \frac{n_i}{\sum_{i=1}^{k} n_i} \tag{5}$$

The input images are pre-processed as below:

1) Resizing

The size of raw images is 784*784, which is so large that the speed of training models is too slow. To avoid losing useful information caused by resizing images too small, the images here are processed to 224*224.

2) Normalization

To help the model converge faster, images are normalized when loaded. More specifically, the average value and variance of images in one dataset are computed beforehand. When reading a single image from a dataset, the value on every pixel of the image is used to subtract the average value, and then divided by the variance.

D. Result Analyses

The model performance is evaluated mainly by the accuracy on the validation set, while ROC, AUC, and confusion matrix are sometimes taken into account.

The results of Experiments A and B are compared with Experiment C in Fig. 4. The values of the average accuracy of the last 75 epochs on the validation set are presented in Table II.

TABLE II. AVERAGE ACCURACY OF ALL CLIENTS IN EXPERIMENT A, B AND C

Experiment	Average accuracy of different clients		
	$\mathcal{D}_1$	$\mathcal{D}_2$	$\mathcal{D}_3$
A	0.7666	0.6879	0.6413
B	0.8206	0.6831	0.6475
C	0.8237	0.7338	0.6691
Locally Training	0.8170	0.4818	0.4410

On client $\mathcal{D}_1$ (Fig. 4(a)), the accuracy of the locally trained model is higher than that of the federated trained model in Experiment A. Because $\mathcal{D}_1$ has the largest data amount (708 images), which is much larger than the data volume of the other two clients, the accuracy of locally trained model on $\mathcal{D}_1$ is much higher than those on $\mathcal{D}_2$ and $\mathcal{D}_3$. Therefore, the global optimum strategy adopted by Experiment A can obtain lower global losses when the aggregated model distills more knowledge from $\mathcal{D}_2$ and $\mathcal{D}_3$, and finally resulted in that the aggregated model performed worse on $\mathcal{D}_1$ than the locally trained model. Therefore, in Experiment B, where the locally optimum strategy is adopted, the average accuracy on $\mathcal{D}_1$ is 0.0036 higher than the accuracy of the locally trained model (see Table II). In other words, the trick of handling *feature heterogeneity* resulted in an accuracy increase of 0.066. On the contrary, it does not work well on $\mathcal{D}_2$ and $\mathcal{D}_3$ with only more than 100 images.

On $\mathcal{D}_2$ and $\mathcal{D}_3$, the locally trained model always has the lowest accuracy (Fig. 4(b) and Fig. 4(c)), indicating that federated learning plays an important role here and brings an accuracy improvement of at least 0.2 (Table II). On the basis of Experiment B with little accuracy promotion, in Experiment C with the weighted averaging strategy, both the accuracy of $\mathcal{D}_2$ and $\mathcal{D}_3$ has a significant increase ($\mathcal{D}_2$: 0.0507, $\mathcal{D}_3$: 0.0216, see Table II.). This is because $\mathcal{D}_1$ has a much

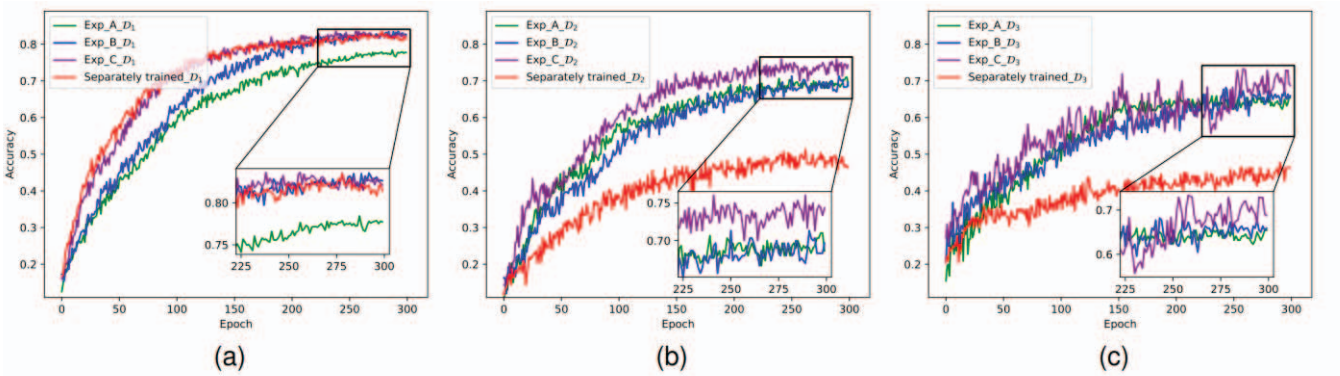

Fig. 4. The averaged Accuracy-Epoch curves on validation set of three clients. Each subfigure is corresponding to a specific client and contains 4 curves, which represent the results of Experiment A, B, C and separately training.

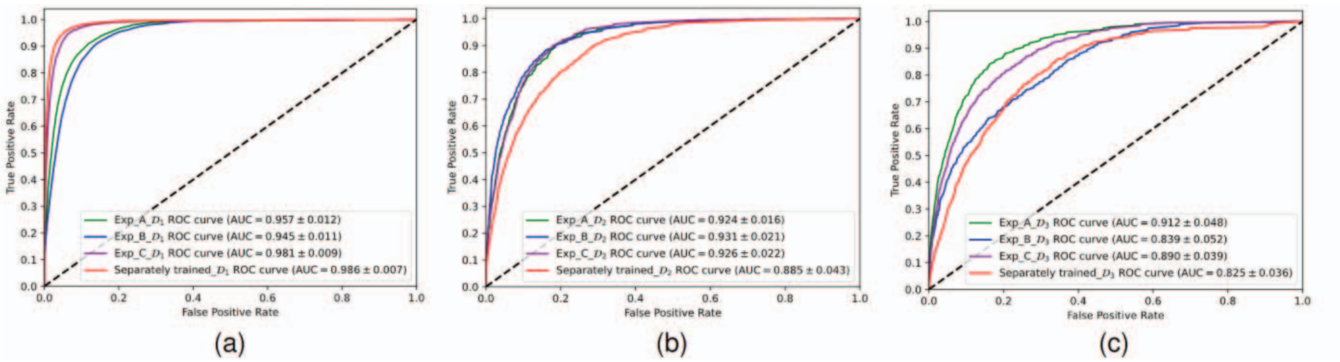

Fig. 2. The micro-averaged ROC curves and corresponding AUC values (with 95% confidence interval) on the dataset of three clients. Each subfigure is related to a client and contains 4 curves, which represent the results of Experiment A, B, C and separately training.

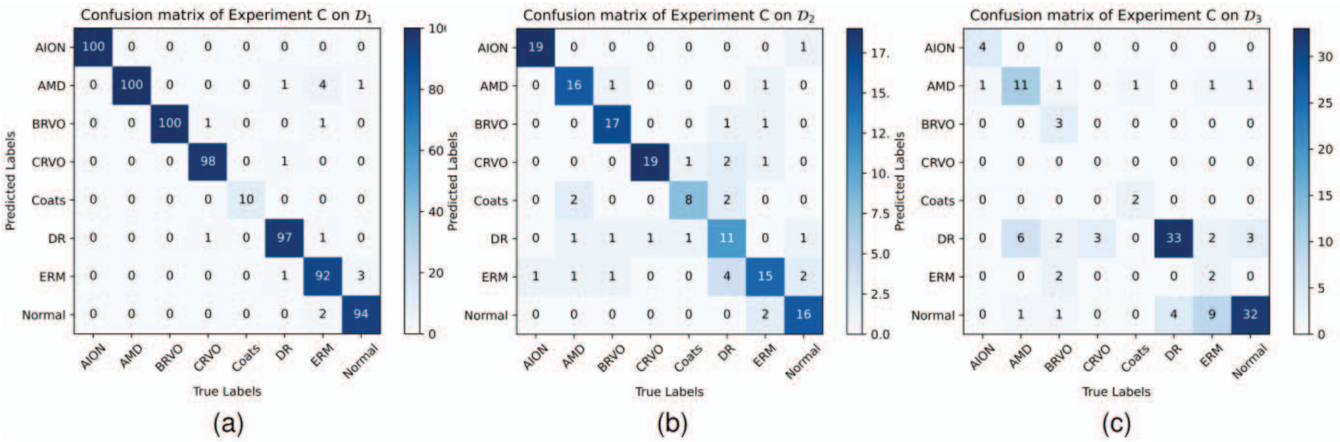

Fig. 7. The confusion matrixes of $\mathcal{D}_1$, $\mathcal{D}_2$ and $\mathcal{D}_3$ in Experiment C, calculated by using models of the best performed training round.

larger data amount, $\mathcal{D}_2$ and $\mathcal{D}_3$ can learn much more knowledge than they can in their own datasets during the progress of federated learning.

The AUC values of experiments A to C for the same client in Fig. 5 do not show significant differences or obvious trends, and these values are larger than 0.900 in most cases, which is enough to illustrate that the federated disease diagnostic model is quite effective. In order to better illustrate the performance of the disease diagnosis model on each category, we took Experiment C as an example and calculated the confusion matrixes using models that achieved the best performance in Experiment C, as shown in Fig. 6. Models corresponding to $\mathcal{D}_1$ and $\mathcal{D}_2$ work very well, while the model corresponding to $\mathcal{D}_3$ has some flaws that can be seen in Fig. 6(c) The model incorrectly classified 36 images from multiple disease categories as Normal, possibly due to the extreme imbalance of data amount between categories in $\mathcal{D}_3$.

Generally speaking, the performance of the model can be significantly improved by using a federated learning paradigm compared with only using local data to train it. Besides, the data amount of the local client is positively correlated to the accuracy of the single trained local model. The combination of techniques for dealing with data amount heterogeneity and feature heterogeneity leads to a significant increase in the effectiveness of federated learning, although the effectiveness of each technique may vary depending on the amount of data on each client.

IV. Conclusion

In this article, we first analyzed the cause of the formation of *data amount heterogeneity and feature heterogeneity* in FFA image datasets and their harm to the performance of the diagnostic model, and then proposed the modeling strategies to handle corresponding data heterogeneity. Weighted averaging was used to tackle *data amount heterogeneity*, and local optimization was used to tackle *feature heterogeneity*. Synthesized these strategies above, we proposed a practical heterogeneous federated learning method for multi-centric ophthalmic disease diagnosis, which finally improved the average diagnostic accuracy of 3 clients by 19.31% compared with locally trained clients. Ablation experiments have validated that the proposed methods have significant improvements, and the average accuracy is 4.36% higher than that of pure federated learning.

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